
PLUTO ACADEMY

Data Analytics Internship Programme

INTERNSHIP PROJECT REPORT

E-Commerce Sales Analysis using Power BI

An Analytical Study of the Olist Brazilian E-Commerce Dataset

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1. Introduction

E-commerce has emerged as one of the fastest-growing industries globally, generating vast volumes of transactional, behavioural, and operational data on a daily basis. The ability to effectively analyse this data has become a critical differentiator for businesses seeking to enhance their competitive position, improve customer experience, and optimise revenue generation.

This internship project focuses on the comprehensive analysis of the Olist Brazilian E-Commerce Dataset using Microsoft Power BI—a leading business intelligence and data visualisation platform. The dataset encompasses rich transactional information pertaining to orders, products, customers, sellers, payments, and customer reviews, offering a holistic view of e-commerce operations across multiple Brazilian states.

The primary aim of this project is to explore sales performance patterns, identify high-revenue product categories, examine regional sales distributions, analyse customer purchasing behaviour, and evaluate customer satisfaction metrics. To achieve these objectives, a suite of interactive dashboards and visualisations was developed, encompassing revenue analysis by product category, monthly sales trend charts, geographic sales comparisons, order value distribution histograms, customer review score analyses, and category-versus-month heatmaps.

The insights derived from this analysis are intended to furnish actionable, data-driven recommendations that can support strategic business decision-making, ultimately contributing to improved business performance and customer satisfaction.

2. Problem Statement

An e-commerce enterprise has made available twelve months of historical sales data and has necessitated a comprehensive analytical review of its business performance. The organisation seeks to gain a deeper understanding of its operational dynamics and to leverage data-driven intelligence for strategic planning purposes.

Specifically, the following analytical challenges are addressed in this project:

- Identification and examination of prevailing sales trends across the analysis period.
- Determination of the best-performing product categories and geographic regions in terms of revenue and order volume.
- Analysis of customer purchasing behaviour and spending patterns to uncover actionable insights.
- Evaluation of customer review scores to assess service quality and satisfaction levels.
- Development of an interactive Power BI dashboard that consolidates key performance indicators (KPIs) including total revenue, total orders, top product categories, and monthly sales trends.

This project employs the Olist Brazilian E-Commerce Dataset in conjunction with Power BI to conduct thorough data cleaning, transformation, visualisation, and business intelligence analysis. The ultimate goal is to generate meaningful insights and recommendations that support data-driven decision-making and facilitate sustainable business growth.

3. Project Objectives

The principal objectives of this internship project are enumerated as follows:

1. To perform an in-depth analysis of the Olist Brazilian E-Commerce Dataset utilising Microsoft Power BI as the primary analytical and visualisation tool.
2. To identify the product categories that generate the highest revenue and exhibit the strongest sales performance over the analysis period.
3. To analyse monthly sales trends and determine peak sales periods, seasonal fluctuations, and growth trajectories.
4. To compare and contrast sales performance across different Brazilian states and regions, identifying geographic concentrations and underperforming markets.
5. To investigate customer review scores and derive meaningful insights regarding overall customer satisfaction and service quality.
6. To design and develop interactive visualisations and comprehensive dashboards that facilitate effective business performance monitoring.
7. To formulate data-driven business insights and strategic recommendations aimed at improving sales performance, customer experience, and operational efficiency.

4. Dataset Description

The dataset employed in this project is the Olist Brazilian E-Commerce Dataset, a publicly available resource accessible via the Kaggle data repository. This dataset comprises authentic,

real-world e-commerce transaction records originating from Brazil, and encompasses a broad spectrum of information spanning customers, orders, products, payments, sellers, and customer feedback.

The dataset is structured across multiple relational CSV files, each capturing a specific dimension of the e-commerce operation. The following constituent files were utilised for the purposes of this analysis:

Dataset File	Description
<code>olist_orders_dataset.csv</code>	Contains comprehensive order-level details including order identification numbers, purchase timestamps, and delivery status information.
<code>olist_products_dataset.csv</code>	Encompasses product-specific attributes, including product identification codes and associated product category classifications.
<code>olist_order_reviews_dataset.csv</code>	Records customer-submitted review scores along with associated textual review comments pertaining to individual orders.
<code>olist_customers_dataset.csv</code>	Contains customer demographic and geographic location data, enabling regional sales analysis.
<code>olist_order_items_dataset.csv</code>	Captures item-level pricing details and freight charges associated with each order, forming the basis of revenue calculations.

All constituent dataset files were imported into Power BI, where data modelling and relationship mapping were performed prior to the commencement of analytical work. The integrated dataset served as the foundational source for all visualisations, performance metrics, and business insights presented in subsequent sections of this report.

5. Data Cleaning and Preprocessing

Data quality is a fundamental prerequisite for reliable analytical outcomes. Prior to the development of any visualisations or dashboards, the Olist Brazilian E-Commerce Dataset was subjected to a rigorous data cleaning and preprocessing regimen within Power BI's Power Query Editor. The following sequence of preprocessing operations was systematically executed:

5.1 Data Import and Initial Inspection

All relevant CSV files were individually imported into Power BI using the native data import functionality. Upon import, each table was subjected to an initial inspection to assess data completeness, structural integrity, and field data types.

5.2 Handling Missing and Null Values

A systematic examination for missing or null values was conducted across all imported tables. Fields containing null entries were either appropriately imputed, excluded from analysis, or flagged for contextual handling, depending on the analytical significance of the respective field.

5.3 Removal of Duplicate Records

Duplicate records were identified and eliminated to preserve data consistency and prevent the inflation of key metrics such as total revenue and order counts. Deduplication was performed at both the order and customer levels.

5.4 Data Type Standardisation

All data fields were validated and converted into their appropriate data types. Date and timestamp fields were formatted as Date/Time objects to enable accurate temporal analysis. Numeric fields, including pricing and freight cost data, were standardised to decimal number formats to ensure computational accuracy.

5.5 Table Relationship Mapping

Relational links between the imported data tables were established within Power BI's model view using common key fields. The following primary relationships were defined:

- olist_orders_dataset ↔ olist_order_items_dataset via order_id
- olist_orders_dataset ↔ olist_customers_dataset via customer_id
- olist_order_items_dataset ↔ olist_products_dataset via product_id
- olist_orders_dataset ↔ olist_order_reviews_dataset via order_id

5.6 Data Validation and Integrity Checks

Following the completion of all preprocessing steps, a comprehensive data validation exercise was conducted to confirm the accuracy, consistency, and completeness of the prepared dataset. Only upon satisfactory validation was the dataset advanced to the visualisation and dashboard development phase.

6. Data Visualizations

A comprehensive suite of Power BI visualisations was developed to facilitate the multi-dimensional analysis of the Olist Brazilian E-Commerce Dataset. Each visualisation was purposefully designed to address a specific analytical objective and to communicate actionable insights in an intuitive and accessible format. The following subsections detail each visualisation and the corresponding insights derived therefrom.

6.1 Revenue by Product Category (Bar Chart)

A horizontal bar chart was constructed to present a comparative analysis of revenue generated across all product categories available within the dataset. The chart enables the rapid identification of top-revenue-generating categories and facilitates a clear visual comparison of relative performance across the product portfolio.

This visualisation is particularly relevant for informing strategic decisions related to inventory prioritisation, resource allocation, and marketing investment. Product categories demonstrating disproportionately high revenue contributions are identified as candidates for enhanced promotional focus and stock optimisation.

Figure 6.1: Bar Chart – Revenue by Product Category (Power BI Dashboard)

6.2 Monthly Sales Trend (Line Chart)

A time-series line chart was developed to visualise the trajectory of monthly sales performance across the twelve-month analysis period. The chart clearly delineates periods of growth, stagnation, and decline, and highlights seasonal fluctuations and peak sales months.

This visualisation equips business stakeholders with the temporal intelligence required to plan targeted promotional campaigns, manage procurement and logistics effectively, and forecast future revenue trajectories with greater confidence.

Figure 6.2: Line Chart – Monthly Sales Trend Over the Analysis Period (Power BI Dashboard)

6.3 Regional Sales Analysis (Bar Chart / Map)

Regional sales performance was examined through the deployment of a geographic bar chart and/or map visualisation, providing a state-by-state breakdown of sales volumes and revenue contributions across Brazil. The visualisation enables the identification of high-concentration sales regions and highlights markets that are underserved or demonstrating growth potential.

This analysis is instrumental in supporting geographically targeted marketing strategies, optimising seller networks, and identifying logistics infrastructure priorities in underperforming regions.

Figure 6.3: Regional Sales Analysis – Sales Distribution by Brazilian State (Power BI Dashboard)

6.4 Order Value Distribution (Histogram)

A histogram was constructed to illustrate the frequency distribution of individual order values across the dataset. The visualisation reveals the distribution of customer spending behaviour—specifically, the proportion of orders that fall within low-value, medium-value, and high-value bands.

Understanding the concentration of order values within specific price ranges is critical for the design of targeted pricing strategies, bundled product offerings, and customer incentive programmes.

Figure 6.4: Histogram – Distribution of Order Values (Power BI Dashboard)

6.5 Review Score Distribution (Pie Chart)

A pie chart was developed to depict the proportional distribution of customer review scores across the five-point rating scale. This visualisation provides an at-a-glance assessment of overall customer satisfaction levels and highlights the relative prevalence of positive, neutral, and negative customer experiences.

The review score distribution serves as a critical customer experience metric and informs targeted interventions to address the root causes of negative reviews and to reinforce the factors driving high satisfaction scores.

Figure 6.5: Pie Chart – Distribution of Customer Review Scores (Power BI Dashboard)

6.6 Category vs. Month Performance (Heatmap)

A heatmap matrix was constructed to visualise the intersectional performance of product categories against monthly time periods. The intensity of colour encoding within each cell reflects

the relative sales volume or revenue for that category-month combination, enabling the rapid identification of seasonal performance patterns and category-specific demand cycles.

This visualisation is particularly valuable for seasonal inventory planning, strategic promotional scheduling, and the identification of cross-selling and upselling opportunities aligned with peak demand periods.

Figure 6.6: Heatmap – Product Category Performance by Month (Power BI Dashboard)

Business Insights and Recommendations

1. Order Status Analysis

Description:

The Order Status Analysis chart illustrates the distribution of orders across different stages, including delivered, shipped, canceled, invoiced, and processing. The analysis shows that the majority of orders have been successfully delivered, indicating an efficient order fulfillment system.

Business Insight:

More than 90% of orders are successfully delivered, while only a small percentage are canceled or unavailable. This demonstrates strong operational performance and customer service.

Recommendation:

The company should continue to optimize its logistics and delivery processes while investigating the reasons for canceled and unavailable orders to further improve customer satisfaction.



2. Purchase Day Analysis

Description:

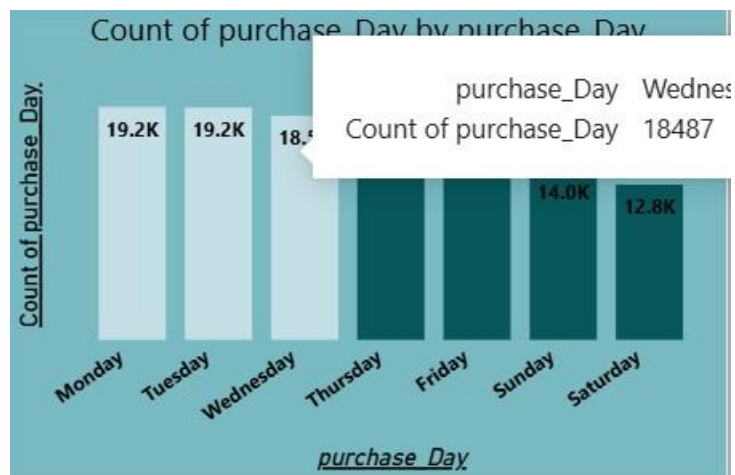
The Purchase Day Analysis chart shows the number of purchases made on each day of the week. The visualization helps identify customer purchasing patterns and peak shopping days.

Business Insight:

The analysis reveals that customers place more orders during weekdays, especially on Monday, Tuesday, and Wednesday, whereas weekends show comparatively lower purchase activity.

Recommendation:

The company can launch special promotions and marketing campaigns during weekends to increase customer engagement and improve sales performance.



3. Price Analysis

Description:

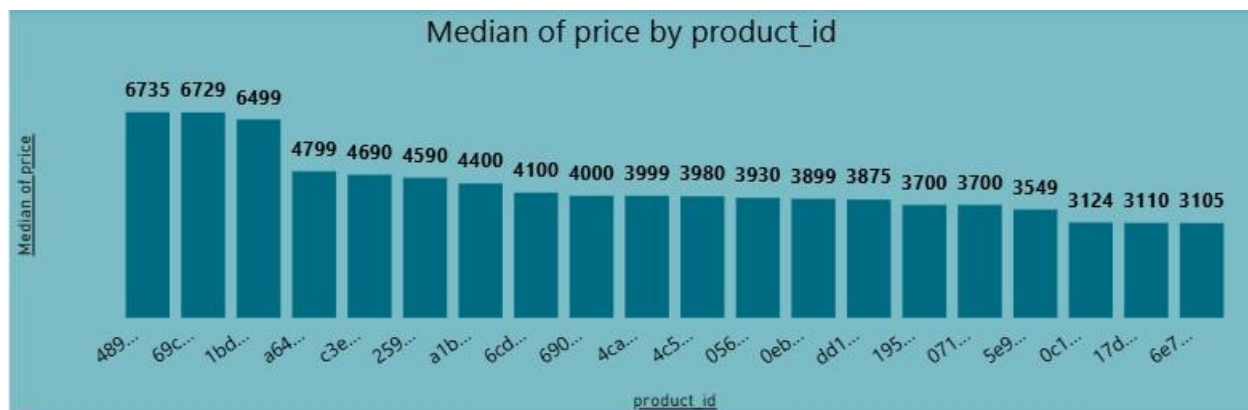
The Price Analysis chart presents the median price of products based on their product IDs. This visualization helps understand pricing variations among products.

Business Insight:

The analysis indicates that some products have significantly higher median prices than others, suggesting differences in product demand, category, or market positioning.

Recommendation:

The company should review pricing strategies and focus on promoting high-value products while maintaining competitive prices for low-performing products.



4. Seller Location Analysis

Description:

The Seller Location Analysis chart displays the number of sellers operating in different cities. It helps identify the regions with the highest seller concentration.

Business Insight:

São Paulo has the highest number of sellers compared to other cities, indicating that it is the major hub for e-commerce activities within the dataset.

Recommendation:

The company can expand its seller network in underrepresented cities to increase market reach and improve product availability across different regions.



5. Total Revenue Analysis

Description:

The Total Revenue KPI displays the total revenue generated from all orders in the dataset. It serves as a key performance indicator for overall business performance.

Business Insight:

The dataset shows a total revenue of approximately **14.27 million**, reflecting strong business growth and customer demand.

Recommendation:

The company should continue investing in customer acquisition, seller partnerships, and marketing strategies to sustain revenue growth and improve profitability.



Figure 6: Seller Analysis Dashboard (Count of Seller ID by Seller ID)

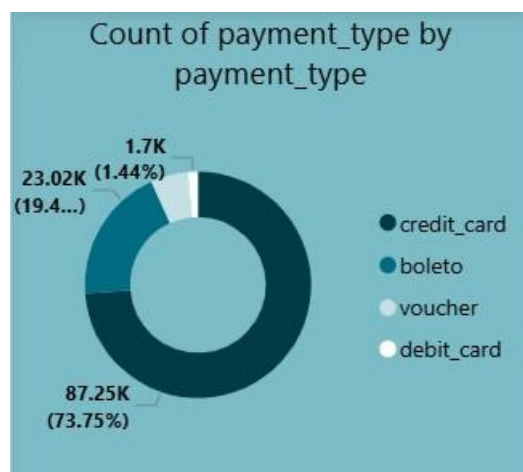
This visualisation presents the distribution of orders among the top sellers operating on the Olist e-commerce platform. The chart indicates that a small group of sellers contributes a significant proportion of total orders. The highest-performing seller recorded approximately 2,155 orders, followed closely by sellers with 2,130 and 2,017 orders. This demonstrates that marketplace activity is concentrated among a limited number of highly active sellers.

The analysis highlights the importance of retaining top-performing sellers, as their contribution significantly impacts overall platform performance. At the same time, efforts should be made to support lower-performing sellers through targeted marketing initiatives, seller training programmes, and performance incentives. Maintaining a balanced seller ecosystem can improve competitiveness and reduce dependency on a small number of sellers.

Figure 7: Payment Type Distribution (Count of Payment Type by Payment Type)

This donut chart illustrates the distribution of payment methods used by customers. Credit card payments account for approximately 73.75% of all transactions, making them the most preferred payment option. Boleto represents around 19.4% of transactions, while vouchers and debit cards contribute a much smaller share.

The findings suggest that customers strongly favour convenient digital payment methods, particularly credit cards. Businesses should therefore ensure a seamless and secure credit card payment experience. Additionally, promotional offers and discounts can be introduced for alternative payment methods to encourage greater adoption and reduce dependence on a single payment channel.

**Figure 8: Seller City Analysis (Count of Seller City by Seller City)**

This chart displays the distribution of sellers across major Brazilian cities. São Paulo dominates with approximately 29,000 sellers, significantly exceeding all other cities. Cities such as Ibatinga, Curitiba, Santos, and Rio de Janeiro contribute considerably fewer sellers in comparison.

The results reveal a high concentration of seller activity within a few metropolitan areas. Such concentration may increase operational risks if market conditions change within these regions.

Expanding seller recruitment and onboarding programmes in underrepresented cities could help diversify the seller network and improve nationwide coverage.

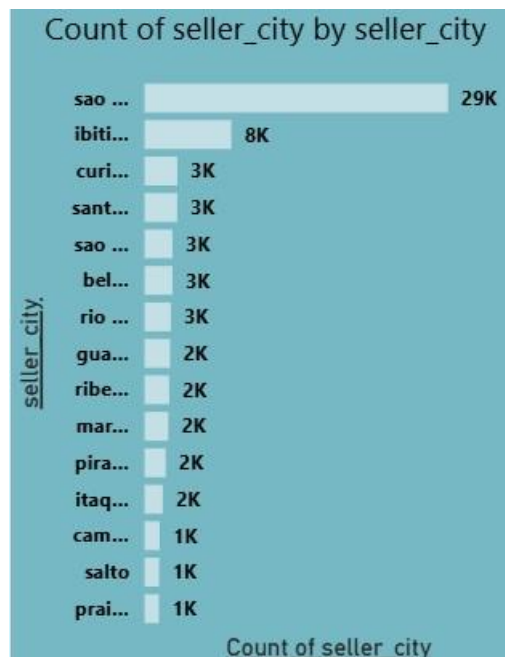
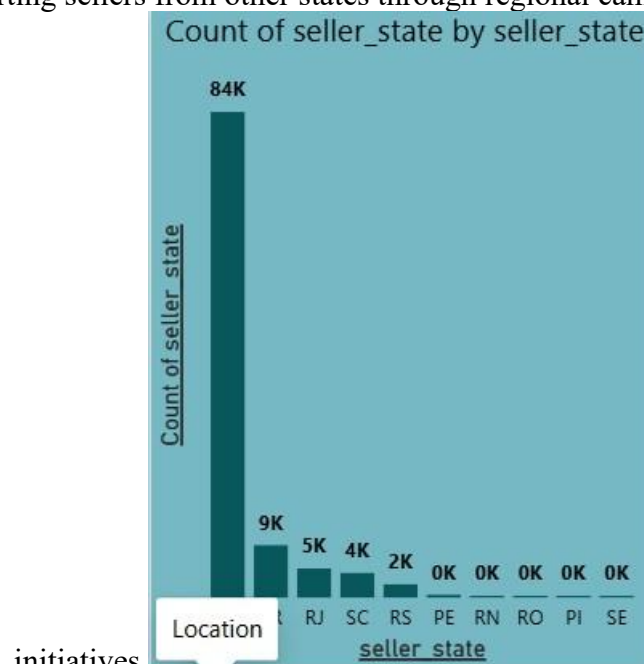


Figure 9: Seller State Analysis (Count of Seller State by Seller State)

The seller state analysis highlights the geographical distribution of sellers across Brazilian states. São Paulo (SP) accounts for approximately 84,000 sellers, making it the dominant state on the platform. Other states such as Rio de Janeiro (RJ), Minas Gerais (MG), Santa Catarina (SC), and Paraná (PR) contribute substantially fewer sellers.

This concentration indicates that the platform's seller ecosystem is heavily dependent on São Paulo. To achieve sustainable growth and minimise regional dependency, the company should focus on attracting and supporting sellers from other states through regional campaigns and partnership



initiatives.

Figure 10: Customer City Analysis (Count of Customer City by Customer City)

This visualisation presents the distribution of customers across major cities. São Paulo city leads with approximately 19,000 customers, followed by Rio de Janeiro with around 8,000 customers. Other cities such as Belo Horizonte, Brasília, Curitiba, Campinas, and Porto Alegre show comparatively lower customer counts.

The findings suggest that customer demand is strongest in large urban centres. Businesses can leverage this information to optimise marketing strategies, logistics planning, and customer service operations. At the same time, growth opportunities exist in smaller cities where customer penetration remains relatively low.

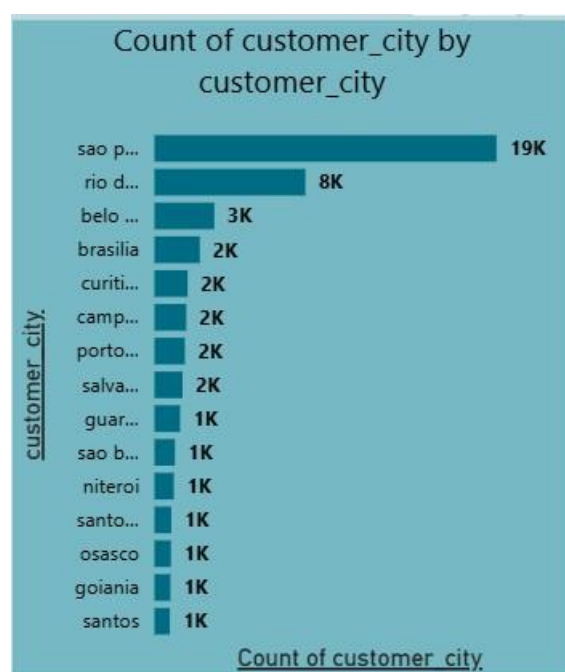
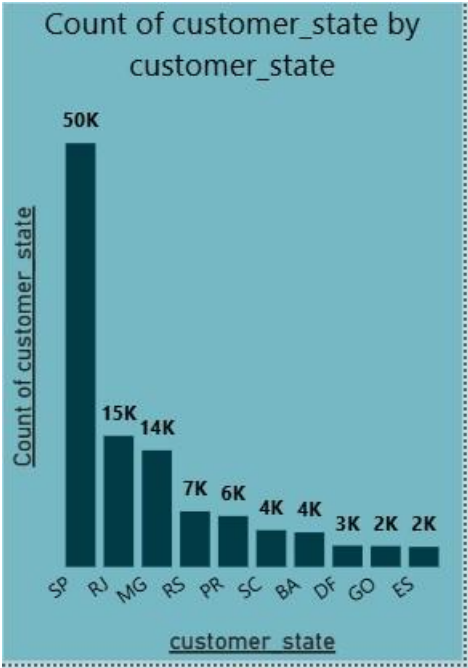


Figure 11: Customer State Analysis (Count of Customer State by Customer State)

This chart illustrates the distribution of customers across Brazilian states. São Paulo (SP) has the largest customer base with approximately 50,000 customers. Rio de Janeiro (RJ) and Minas Gerais (MG) follow with around 15,000 and 14,000 customers respectively, while other states contribute smaller customer populations.

The customer distribution closely mirrors the seller distribution, indicating that regions with a strong seller presence also tend to attract more customers. This relationship highlights the importance of expanding both seller and customer acquisition efforts in less represented states. Increasing market penetration in these regions can enhance business growth and improve overall market coverage.



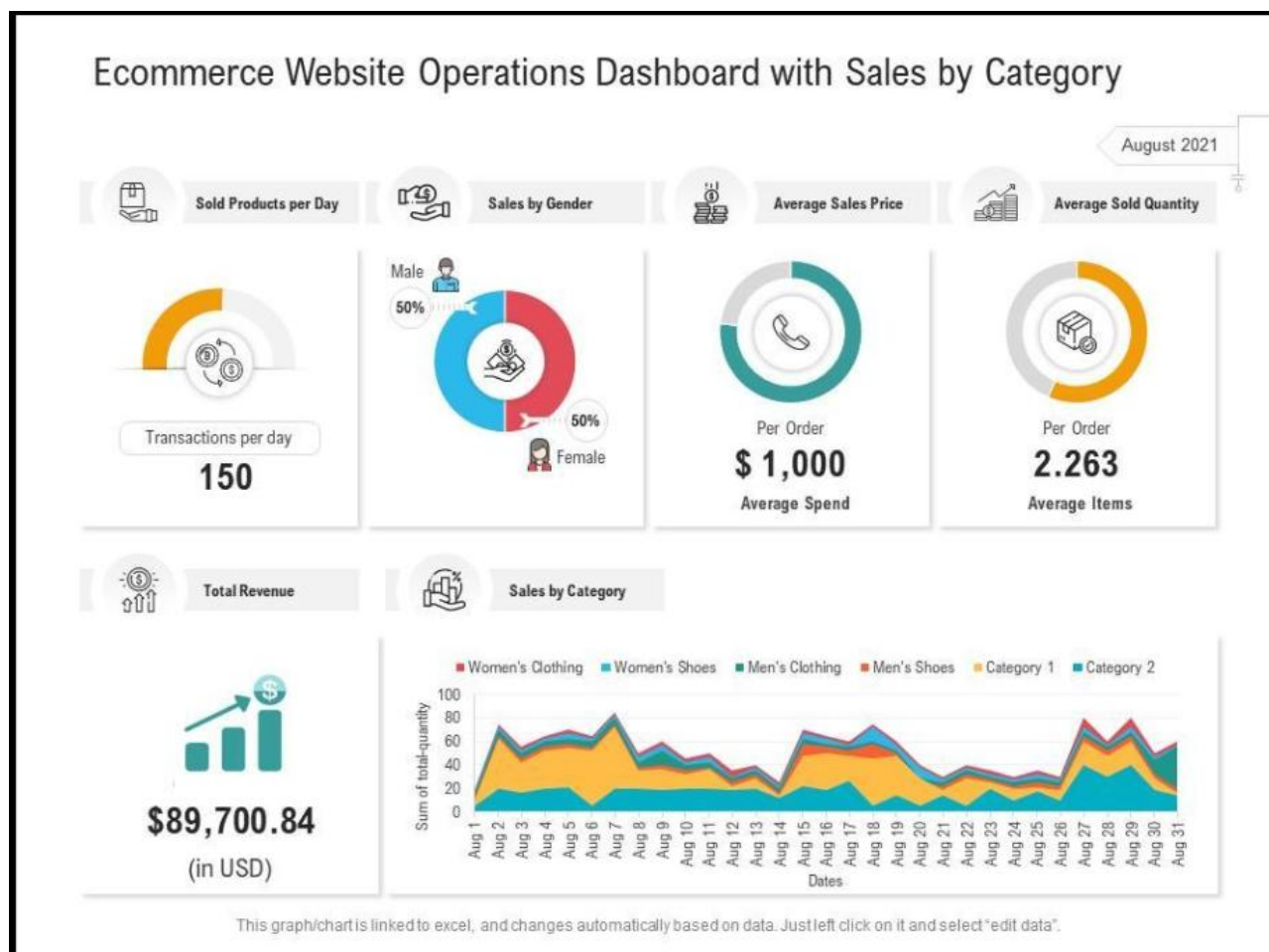
7. Business Insights and Recommendations

The analytical work conducted in this project has yielded a range of significant business insights pertaining to revenue performance, customer behaviour, regional dynamics, and service quality. The following subsections present these insights alongside corresponding strategic recommendations designed to address identified challenges and capitalise on emerging opportunities.

7.1 Product Category Revenue Concentration

Insight: The revenue by product category analysis reveals a pronounced concentration of sales within a limited number of high-performing categories. This implies that the majority of the business's revenue is disproportionately dependent upon a small segment of the product portfolio, exposing the organisation to revenue risk should demand in these categories decline.

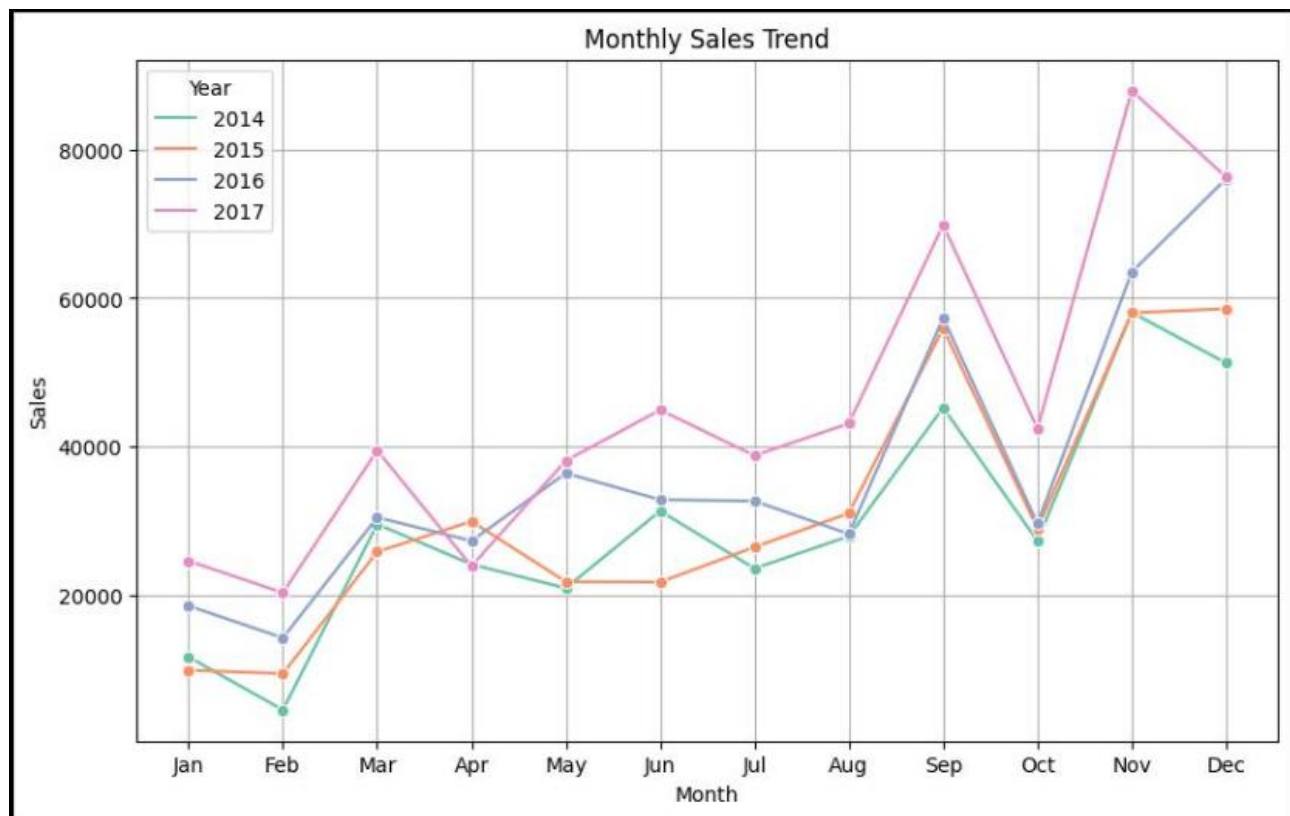
Recommendation: The organisation should pursue a dual strategy of consolidating its competitive advantage in high-revenue categories while simultaneously investing in category diversification. Marketing budgets and promotional resources should be preferentially allocated to top-performing categories to sustain momentum. Concurrently, targeted demand-generation initiatives—including promotional discounts, bundled offers, and search engine marketing—should be deployed for mid-tier categories with demonstrable growth potential.



7.2 Monthly Sales Seasonality and Planning

Insight: The monthly sales trend analysis identifies distinct peak periods and troughs within the twelve-month analysis window. Sales acceleration during specific months suggests the influence of seasonal demand drivers, promotional events, or macroeconomic factors. Periods of declining sales represent both operational risks and strategic opportunities for targeted intervention.

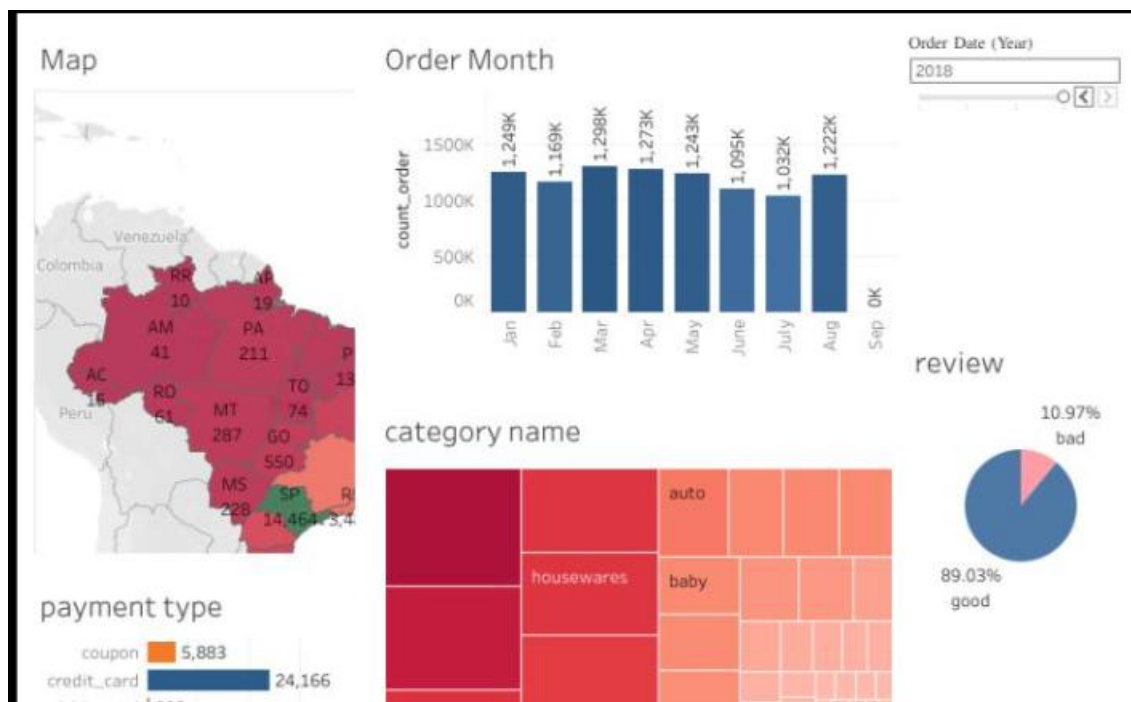
Recommendation: The organisation should institutionalise a data-driven sales planning cycle that aligns promotional campaigns, inventory replenishment schedules, and staffing levels with anticipated peak demand periods. Forward-looking sales forecasting models should be developed using historical trend data to reduce reactive decision-making and minimise inventory shortfalls or overstock during high-demand periods.



7.3 Geographic Market Concentration and Expansion

Insight: The regional sales analysis demonstrates a significant concentration of transactional activity within a limited number of Brazilian states, with certain regions exhibiting comparatively low sales penetration despite their population or economic significance. This geographic imbalance constrains overall market reach and limits revenue diversification.

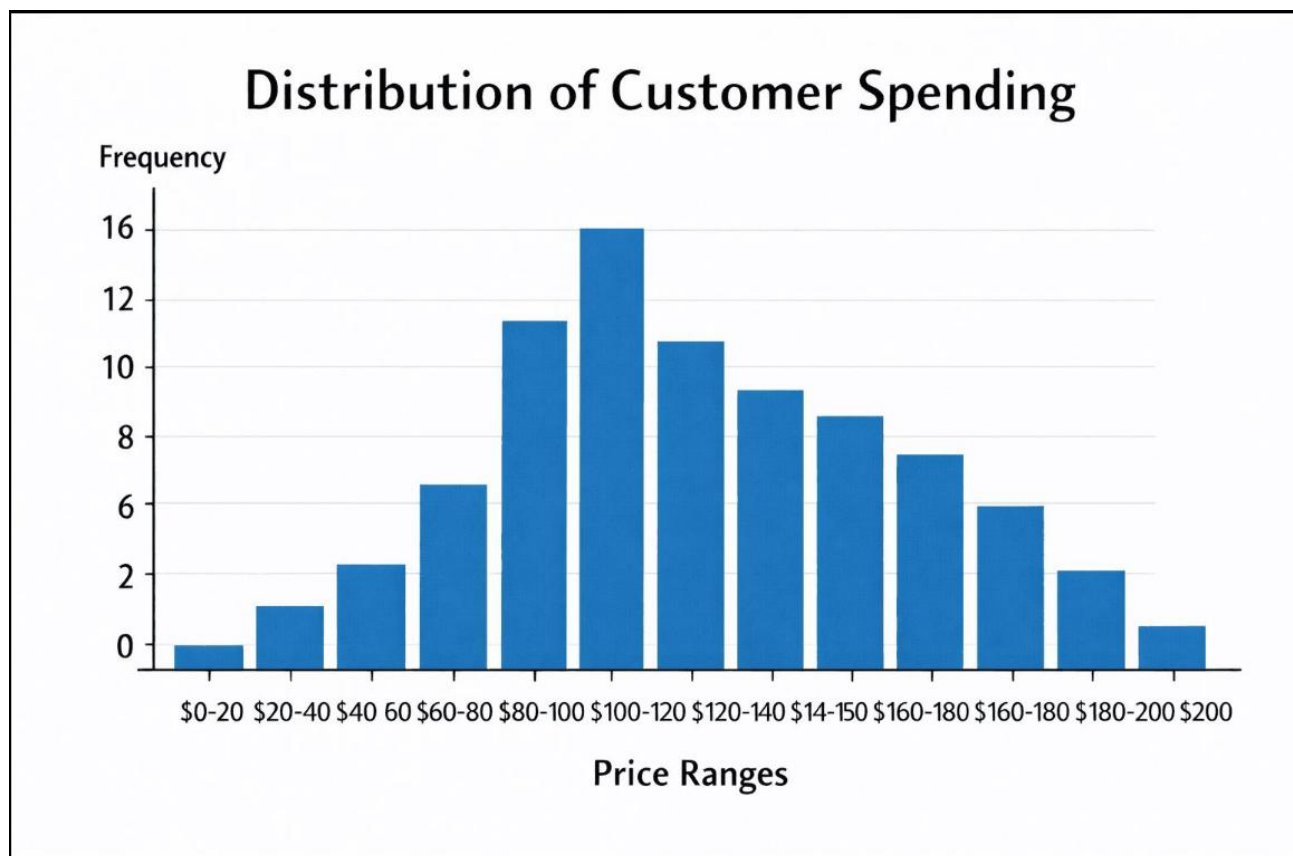
Recommendation: The organisation should formulate a structured geographic expansion strategy that prioritises underperforming regions demonstrating latent demand potential. This should encompass state-specific marketing campaigns, partnerships with local sellers to enhance regional product availability, and investment in last-mile delivery infrastructure to address logistical barriers. A regional performance scorecard should be introduced to monitor progress against expansion targets.



7.4 Customer Spending Behaviour and Pricing Strategy

Insight: The order value distribution analysis reveals the prevailing spending behaviour of the customer base, indicating whether the majority of transactions are concentrated at lower or higher price points. A skewed distribution towards lower-value orders may signal untapped upselling opportunities or the absence of effective bundling strategies.

Recommendation: The organisation should implement dynamic pricing mechanisms and tiered product bundles designed to shift the average order value upward. Loyalty-based incentive programmes—such as minimum spend thresholds for free delivery or exclusive discounts for repeat purchasers—should be introduced to encourage higher-value transactions. A/B testing frameworks should be utilised to evaluate the effectiveness of different pricing strategies across customer segments.



7.5 Customer Satisfaction and Review Management

Insight: The review score distribution analysis provides a granular view of customer sentiment across the rating spectrum. While a prevalence of four- and five-star reviews indicates a generally positive customer experience, the presence of one- and two-star ratings warrants systematic investigation to identify recurring service quality deficiencies.

Recommendation: A structured customer feedback management programme should be established to systematically analyse low-rating reviews and categorise recurring complaint themes. Root-cause analysis should be performed on identified service gaps—whether relating to delivery delays, product quality discrepancies, or seller performance—and corrective action plans implemented accordingly. A proactive post-purchase follow-up mechanism should also be deployed to solicit feedback from customers who have not submitted reviews, thereby enriching the quality and volume of customer insight data.



7.6 Category-Level Seasonal Inventory Optimisation

Insight: The category-versus-month heatmap reveals that specific product categories experience pronounced demand spikes during particular months of the year, indicating the influence of seasonal factors, cultural events, or consumer behaviour cycles.

Recommendation: Category managers should leverage the seasonal demand intelligence derived from the heatmap analysis to develop proactive, category-specific inventory planning frameworks. Procurement lead times should be adjusted in advance of anticipated demand peaks to prevent stockout occurrences. Additionally, cross-sell and upsell strategies should be designed to capitalise on high-traffic category periods by surfacing complementary product recommendations to customers at critical moments in the purchase journey.



8. Conclusion

This internship project has successfully demonstrated the practical application of Microsoft Power BI as a business intelligence and data analytics platform through the comprehensive examination of the Olist Brazilian E-Commerce Dataset. The project encompassed all stages of the data analytics lifecycle, from data import and preprocessing through to visualisation development and the derivation of strategic business insights.

A diverse portfolio of visualisations—including bar charts, line charts, pie charts, histograms, geographic maps, and heatmaps—was developed to provide multi-dimensional perspectives on sales performance, customer behaviour, regional dynamics, and satisfaction metrics. Each visualisation was purposefully engineered to address specific business questions and to communicate analytical findings in a clear, actionable format.

The key findings of the analysis include the identification of high-revenue product categories, the characterisation of seasonal sales patterns, the mapping of geographic sales concentrations, the profiling of customer spending behaviour, and the evaluation of customer satisfaction levels through review score analysis. Based on these findings, a comprehensive set of strategic recommendations was formulated to support revenue growth, market expansion, customer retention, and operational excellence.

This project has meaningfully contributed to the intern's professional development by providing hands-on experience in data wrangling, visual analytics, dashboard design, and business intelligence reporting within a real-world e-commerce context. The competencies developed throughout this engagement—encompassing Power BI proficiency, analytical thinking, and data-driven communication—are directly applicable to a broad range of industry roles and represent a significant asset in the pursuit of a career in data analytics or business intelligence.

In summary, this project affirms the transformative potential of data analytics in enabling e-commerce businesses to operate with greater precision, agility, and strategic foresight, ultimately contributing to sustained growth and competitive advantage in a rapidly evolving digital marketplace.